

MoonBit Typographical Compiler

Architecture & Features

Slide 1 Introduction to moon-typst

moon-typst brings modern typographical typesetting to the MoonBit ecosystem.

- â€¢ **Zero-FFI Pure MoonBit** 100% portable across Native JS and WebAssembly targets
- â€¢ **Sub-millisecond Compilation**
Incremental layout engine built for interactive documents
- â€¢ **Multi-Target Rendering** Seamless output to SVG semantic HTML5 and standard PDF 1.4
- â€¢ **Rich Mathematical Layout** First-class support for fractions square roots matrices and limits

Slide 2 Mathematical Capabilities

Typesetting complex scientific formulas with pixel-perfect alignment

$$\frac{\alpha + \beta}{\gamma + \delta}$$

Multivariate transformations with matrix layouts

State vectors in 3D projection

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

Slide 3 Architectural Pipeline

The internal pipeline consists of four modular packages

- â€¢ **core** Abstract syntax tree color representations and length unit conversions
- â€¢ **diag** Source span tracking diagnostic error collection and colored terminal reports
- â€¢ **eval** Lexical scoping variable environments and cascading style rules
- â€¢ **parser** Fast recursive descent lexer and token stream analyzer
- â€¢ **layout** Spatial bounding box calculations text measurement and table layout
- â€¢ **render** SVG vector emitter HTML5 document generator and PDF 1.4 binary serializer

Slide 4 Getting Started

Compile any Typst document directly using the CLI

```
# Render to vector SVG
typst-render presentation.typ -o presentation.svg

# Render to standard PDF 1.4
typst-render presentation.typ -p -o presentation.pdf

# Render to semantic HTML5
typst-render presentation.typ --html -o presentation.html
```

Empowering next-generation cloud and WebAssembly document authoring!